

ANALYYSI I: LIMITS

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ABSTRACT. Notes for the University of Helsinki course MAT11003, taught in Finnish from 01.08.2026–17.10.2026.

CONTENTS

1.	Course information	1
2.	Lecture notes	1
2.1.	01.09.2026	1
2.2.	03.09.2026	2
2.3.	08.09.2026	4
2.4.	10.09.2026	6
2.5.	15.09.2026	10
2.6.	17.09.2026	11
2.7.	22.09.2026	14
2.8.	24.09.2026	15
2.9.	29.09.2026	17
2.10.	01.10.2026	19
2.11.	06.10.2026	24

1. COURSE INFORMATION

The official course information is available on the University of Helsinki course page.

2. LECTURE NOTES

2.1. 01.09.2026.

2.1.1. *The absolute value function.*

Definition 2.1. For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of x , denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points x and y on the real line.

Example 2.2. Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x - y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y - h, y + h). \end{aligned}$$

Thus, $|x - y| < h$ means that x lies in the open interval of radius h centred at y .

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Example 2.3. As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x - 1| < 3 &\iff -3 < 2x - 1 < 3 \\ &\iff -2 < 2x < 4 \\ &\iff -1 < x < 2 \\ &\iff x \in (-1, 2). \end{aligned}$$

Theorem 2.4 (Triangle and reverse triangle inequalities). *For all $x, y \in \mathbb{R}$,*

$$\begin{aligned} |x + y| &\leq |x| + |y|, \\ ||x| - |y|| &\leq |x - y|. \end{aligned}$$

Proof. By definition 2.1,

$$\begin{aligned} -|x| &\leq x \leq |x|, \\ -|y| &\leq y \leq |y|. \end{aligned}$$

Adding the inequalities gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

which is equivalent to $|x + y| \leq |x| + |y|$.

Applying this triangle inequality to $x = (x - y) + y$ gives

$$|x| - |y| \leq |x - y|.$$

Interchanging x and y similarly gives $|y| - |x| \leq |x - y|$. Therefore

$$-|x - y| \leq |x| - |y| \leq |x - y|,$$

which is equivalent to the reverse triangle inequality. □

2.2. 03.09.2026.

2.2.1. More on absolute values.

Theorem 2.5. *We have*

$$|xy| = |x||y|$$

for all $x, y \in \mathbb{R}$.

Proof. We prove by splitting into four cases. Firstly, if $x, y \geq 0$, we have $xy \geq 0$, so $|xy| = xy$. But in this case also both $|x| = x$ and $|y| = y$, so $|xy| = xy = |x||y|$. Secondly, if $x, y < 0$, we have $xy > 0$, $|xy| = xy = (-x)(-y) = |x||y|$. Now suppose $x < 0$ and $y \geq 0$. Then $xy \leq 0$, and hence $|xy| = -xy = (-x)y = |x||y|$. The case $x \geq 0$, $y < 0$ follows from commutativity of multiplication of real numbers. □

Example 2.6. Let $x \in (1, 3)$ and $y \in (4, 8)$. Find an upper bound for

$$|x + y - 8|.$$

Kolmioepäyhtälön nojalla saadaan

$$\begin{aligned} |x + y - 8| &= |(x - 2) + (y - 6)| \\ &\leq |x - 2| + |y - 6| \\ &< 1 + 2 \\ &= 3. \end{aligned}$$

Thus we have found that 3 is an upper bound for $|x + y - 8|$.

Example 2.7. Let $x \in (2, 6)$ and $y \in (2, 8)$. Find an upper bound for

$$|xy - 20|.$$

We have $|x - 4| < 2$ and $|y - 5| < 3$. We have

$$\begin{aligned} |xy - 20| &= |xy - 4y + 4y - 20| \\ &\leq |xy - 4y| + |4y - 20| \\ &= |y||x - 4| + 4|y - 5| \\ &\leq 8 \cdot 2 + 4 \cdot 3 \\ &= 28. \end{aligned}$$

It follows that 28 is an upper bound for $|xy - 20|$.

Example 2.8. Find a positive real number m such that

$$|x^2 - 16| \leq m|x - 4|,$$

where $x \in (0, 10)$.

When $x = 4$, we get $0 \leq 0$, which holds with every $m \in \mathbb{R}$. Now suppose $x \neq 4$. We simplify the inequality and obtain

$$|x + 4| \leq m.$$

Since $x \in (0, 10)$, $|x + 4| \in (4, 14)$. We have thus obtained

$$|x + 4| \leq 14,$$

so we can choose $m = 14$.

2.2.2. Field and ordering properties of $\mathbb{R}$.

Definition 2.9. $(F, +, \cdot)$ is a field if and only if

- (1) The algebraic structure $(F, +)$ is an *abelian group*.
- (2) The algebraic structure $(F^*, \cdot)$ is an *abelian group*, where $F^* = F \setminus \{0_F\}$.
- (3) The operation $\cdot$ distributes over $+$.

Theorem 2.10. $\mathbb{R}$ is a field.

Proof. Consider the usual addition and multiplication on $\mathbb{R}$.

First, $(\mathbb{R}, +)$ is an abelian group. Addition is closed, associative, and commutative; 0 is an additive identity; and every $x \in \mathbb{R}$ has the additive inverse $-x$.

Next, $(\mathbb{R} \setminus \{0\}, \cdot)$ is an abelian group. Multiplication is associative and commutative, and 1 is its identity element. If $x \neq 0$, then $x^{-1} = 1/x$ is a nonzero real number satisfying

$$x \cdot x^{-1} = 1.$$

The product of two nonzero real numbers is nonzero, so multiplication is closed on $\mathbb{R} \setminus \{0\}$.

Finally, multiplication distributes over addition:

$$x(y + z) = xy + xz$$

for all $x, y, z \in \mathbb{R}$. Therefore $(\mathbb{R}, +, \cdot)$ is a field. □

Definition 2.11 (Ordered field). An *ordered field* is a field $(F, +, \cdot)$ equipped with a relation $<$ such that, for all $x, y, z \in F$,

- (1) exactly one of $x < y$, $x = y$, or $x > y$ holds;
- (2) if $x < y$ and $y < z$, then $x < z$;
- (3) if $x < y$, then $x + z < y + z$; and
- (4) if $0_F < x$ and $0_F < y$, then $0_F < xy$.

Here $x > y$ means $y < x$. The first two conditions state that $<$ is a strict total order, while the last two state that the order is compatible with the field operations.

Theorem 2.12. *The field $(\mathbb{R}, +, \cdot)$, equipped with the usual order $<$, is an ordered field.*

Proof. We have already shown that $(\mathbb{R}, +, \cdot)$ is a field. The usual strict order on $\mathbb{R}$ satisfies trichotomy and transitivity, so it satisfies the first two ordering axioms.

Suppose that $x < y$. Then $0 < y - x$, and for every $z \in \mathbb{R}$,

$$(y + z) - (x + z) = y - x > 0.$$

Hence $x + z < y + z$. Furthermore, if $0 < x$ and $0 < y$, then the usual sign rule for multiplication gives $0 < xy$. Thus the order is compatible with addition and multiplication, and $\mathbb{R}$ is an ordered field. $\square$

Theorem 2.13 (Uniqueness of additive identity). *The identity element 0_F of the abelian group $(F, +)$ is unique.*

Proof. Suppose that 0_F and $0'_F$ are both additive identity elements of $(F, +)$. Since 0_F is an additive identity,

$$0_F + 0'_F = 0'_F.$$

Since $0'_F$ is also an additive identity,

$$0_F + 0'_F = 0_F.$$

Therefore $0_F = 0'_F$. $\square$

Theorem 2.14 (Cancellation law). *Let $x, y, z \in \mathbb{R}$. If $x + y = x + z$, then $y = z$.*

Proof. Using the additive identity, additive inverse, associativity, and the assumption $x + y = x + z$, we obtain

$$\begin{aligned} y &= 0 + y \\ &= ((-x) + x) + y \\ &= (-x) + (x + y) \\ &= (-x) + (x + z) \\ &= ((-x) + x) + z \\ &= 0 + z \\ &= z. \end{aligned}$$

$\square$

2.3. 08.09.2026.

2.3.1. Sequences.

Definition 2.15. Let X and Y be sets. A function $f: X \rightarrow Y$ is a relation $f \subseteq X \times Y$ such that, for every $x \in X$, there exists exactly one $y \in Y$ with $(x, y) \in f$. This unique y is denoted by $f(x)$.

The set X is the *domain* of f , and Y is its *codomain*. The set

$$f(X) = \{f(x) : x \in X\}$$

is its *range* or *image*.

Definition 2.16. A sequence of real numbers is a function $x: \mathbb{N} \rightarrow \mathbb{R}$. For $n \in \mathbb{N}$, the value $x(n)$, usually denoted by x_n , is the n -th term of the sequence. We denote the sequence by $(x_n)_{n \in \mathbb{N}}$, or simply by (x_n) .

2.3.2. Limit of a sequence.

Definition 2.17. Let (x_n) be a sequence of real numbers. We say that (x_n) *converges* to $a \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ with $n > N$,

$$|x_n - a| < \varepsilon.$$

In this case, we write $x_n \rightarrow a$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = a$.

Let us look at some examples.

Example 2.18. Let (x_n) be defined by

$$x_n = \frac{2n}{n+1}.$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 2.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. Now for every $n > N$, we have

$$\begin{aligned} |x_n - 2| &= \left| \frac{2n}{n+1} - 2 \right| \\ &= \frac{2}{n+1} < \frac{2}{n} < \frac{2}{\frac{2}{\varepsilon}} = \varepsilon. \end{aligned}$$

Thus $x_n \rightarrow 2$ when $n \rightarrow \infty$.

Example 2.19. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_7 = 42, \\ x_n &= 7 + \frac{1}{n}, \quad n \geq 8. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 7.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max \left\{ 8, \frac{1}{\varepsilon} \right\}$. Now for every $n > N$, we have

$$|x_n - 7| = \frac{1}{n} < \varepsilon.$$

Thus $x_n \rightarrow 7$ as $n \rightarrow \infty$.

Let us now tackle a simple divergence proof.

Example 2.20. Let us show that the sequence x_n defined by $x_n = n$ diverges. Assume that the sequence converged to some $a \in \mathbb{R}$. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - a| < \varepsilon = 1.$$

Choose $n \in \mathbb{N}$ such that

$$n > \max \{N, |a| + 1\}.$$

By the triangle inequality,

$$n = |n| = |(n - a) + a| \leq |n - a| + |a|.$$

Subtracting $|a|$ from both sides and using $n > |a| + 1$, we obtain

$$|x_n - a| = |n - a| \geq n - |a| > 1,$$

contradicting the assumption $|x_n - a| < 1$. Therefore (x_n) diverges.

Example 2.21. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_{42} = 1, \\ x_n &= \frac{n}{n-42}, \quad n \geq 43. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 1.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$N > \max \left\{ 84, \frac{84}{\varepsilon} \right\}.$$

Now, for every $n > N$, we have $n > 84$, and hence $\frac{n}{2} - 42 > 0$. Therefore

$$\begin{aligned} |x_n - 1| &= \left| \frac{n}{n-42} - 1 \right| \\ &= \frac{42}{n-42} \\ &= \frac{42}{\frac{n}{2} + \frac{n}{2} - 42} \\ &< \frac{84}{n} < \varepsilon. \end{aligned}$$

Thus $x_n \rightarrow 1$ as $n \rightarrow \infty$.

2.4. 10.09.2026.

2.4.1. Uniqueness of limits.

Theorem 2.22 (Uniqueness of limits). *A convergent sequence of real numbers has a unique limit. That is, if $x_n \rightarrow a$ and $x_n \rightarrow b$ as $n \rightarrow \infty$, then $a = b$.*

Proof. Suppose for contradiction that $a \neq b$, and set

$$\varepsilon = \frac{|a - b|}{2} > 0.$$

Since $x_n \rightarrow a$ and $x_n \rightarrow b$, there exist $N_a, N_b \in \mathbb{N}$ such that

$$n > N_a \implies |x_n - a| < \varepsilon$$

and

$$n > N_b \implies |x_n - b| < \varepsilon.$$

Choose $n > \max \{N_a, N_b\}$. By the triangle inequality,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon = |a - b|,$$

which is impossible. Therefore $a = b$. □

Theorem 2.23 (Convergent sequences are bounded). *Every convergent sequence of real numbers is bounded.*

Proof. Suppose that $x_n \rightarrow a$. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < 1.$$

By the triangle inequality, for every $n > N$,

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

Let

$$M = \max \{|x_0|, |x_1|, \dots, |x_N|, 1 + |a|\}.$$

Then $|x_n| \leq M$ for every $n \in \mathbb{N}$, so (x_n) is bounded. □

2.4.2. Algebra of limits.

Theorem 2.24 (Algebraic limit laws). *Suppose that $x_n \rightarrow a$ and $y_n \rightarrow b$, and let $c \in \mathbb{R}$. Then*

- (1) $x_n \pm y_n \rightarrow a \pm b$.
- (2) $cx_n \rightarrow ca$.
- (3) $x_n y_n \rightarrow ab$.
- (4) If $b \neq 0$, then $y_n \neq 0$ for all sufficiently large n , and

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

Proof. (1) Let $\varepsilon > 0$. Since $x_n \rightarrow a$ and $y_n \rightarrow b$, there exist $N_x, N_y \in \mathbb{N}$ such that

$$n > N_x \implies |x_n - a| < \frac{\varepsilon}{2}$$

and

$$n > N_y \implies |y_n - b| < \frac{\varepsilon}{2}.$$

Thus, for $n > \max\{N_x, N_y\}$,

$$|(x_n + y_n) - (a + b)| \leq |x_n - a| + |y_n - b| < \varepsilon.$$

Hence $x_n + y_n \rightarrow a + b$. Replacing y_n and b by $-y_n$ and $-b$ gives $x_n - y_n \rightarrow a - b$.

- (2) If $c = 0$, then $cx_n = ca = 0$ for every n . If $c \neq 0$, choose $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < \frac{\varepsilon}{|c|}.$$

Then, for every $n > N$,

$$|cx_n - ca| = |c||x_n - a| < \varepsilon.$$

Therefore $cx_n \rightarrow ca$.

- (3) Since (y_n) converges, it is bounded. Choose $M > 0$ such that $|y_n| \leq M$ for every n . There exist $N_1, N_2 \in \mathbb{N}$ such that

$$n > N_1 \implies |x_n - a| < \frac{\varepsilon}{2M}$$

and

$$n > N_2 \implies |y_n - b| < \frac{\varepsilon}{2(|a| + 1)}.$$

For $n > \max\{N_1, N_2\}$,

$$\begin{aligned} |x_n y_n - ab| &\leq |x_n y_n - a y_n| + |a y_n - ab| \\ &= |y_n| |x_n - a| + |a| |y_n - b| \\ &< M \frac{\varepsilon}{2M} + |a| \frac{\varepsilon}{2(|a| + 1)} \\ &< \varepsilon. \end{aligned}$$

Therefore $x_n y_n \rightarrow ab$.

- (4) Suppose that $b \neq 0$. Since $y_n \rightarrow b$, there exists $N_0 \in \mathbb{N}$ such that

$$n > N_0 \implies |y_n - b| < \frac{|b|}{2}.$$

The reverse triangle inequality then gives

$$|y_n| \geq |b| - |y_n - b| > \frac{|b|}{2} > 0$$

for every $n > N_0$. In particular, $y_n \neq 0$ eventually. Choose $N_1 \in \mathbb{N}$ such that

$$n > N_1 \implies |y_n - b| < \frac{\varepsilon|b|^2}{2}.$$

For $n > \max\{N_0, N_1\}$,

$$\left| \frac{1}{y_n} - \frac{1}{b} \right| = \frac{|y_n - b|}{|y_n||b|} < \frac{2}{|b|^2} |y_n - b| < \varepsilon.$$

Hence $1/y_n \rightarrow 1/b$; the finitely many terms with $n \leq N_0$ may be defined arbitrarily because they do not affect the limit. Applying the product law to (x_n) and $(1/y_n)$ yields

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

□

2.4.3. Limits and order.

Theorem 2.25. *Suppose that $x_n \rightarrow x$. Then*

- (1) *if $x_n \leq M$ for every $n \in \mathbb{N}$, then $x \leq M$;*
- (2) *if $x_n \geq m$ for every $n \in \mathbb{N}$, then $x \geq m$.*

Proof. (1) Suppose for contradiction that $x > M$, and set

$$\varepsilon = \frac{x - M}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \varepsilon.$$

It follows that

$$x_n > x - \varepsilon = x - \frac{x - M}{2} = \frac{x + M}{2} > M,$$

contradicting $x_n \leq M$. Therefore $x \leq M$.

- (2) Suppose for contradiction that $x < m$, and set

$$\varepsilon = \frac{m - x}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \varepsilon.$$

It follows that

$$x_n < x + \varepsilon = x + \frac{m - x}{2} = \frac{x + m}{2} < m,$$

contradicting $x_n \geq m$. Therefore $x \geq m$.

□

Theorem 2.26 (Squeeze theorem). *Suppose that*

$$x_n \leq y_n \leq z_n$$

for every $n \in \mathbb{N}$. If $x_n \rightarrow a$ and $z_n \rightarrow a$, then $y_n \rightarrow a$.

Proof. Let $\varepsilon > 0$. Since $x_n \rightarrow a$, there exists $N_x \in \mathbb{N}$ such that

$$n > N_x \implies a - \varepsilon < x_n < a + \varepsilon.$$

Similarly, since $z_n \rightarrow a$, there exists $N_z \in \mathbb{N}$ such that

$$n > N_z \implies a - \varepsilon < z_n < a + \varepsilon.$$

Therefore, for every $n > \max \{N_x, N_z\}$,

$$a - \varepsilon < x_n \leq y_n \leq z_n < a + \varepsilon.$$

Hence $|y_n - a| < \varepsilon$, so $y_n \rightarrow a$. $\square$

The inequalities in the squeeze theorem need only hold for all sufficiently large n , since changing finitely many terms does not affect convergence.

Example 2.27. Consider the oscillating sequence

$$x_n = \frac{(-1)^n}{n}, \quad n \geq 1.$$

Its sign alternates, but

$$-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}.$$

Both bounding sequences converge to 0. Indeed, given $\varepsilon > 0$, choosing $N > 1/\varepsilon$ gives $1/n < \varepsilon$ whenever $n > N$. The squeeze theorem therefore gives

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0.$$

Notice that the squeeze theorem lets us prove convergence without trying to handle even and odd indices separately.

Example 2.28. Consider

$$x_n = \sqrt{n^2 + n} - n, \quad n \geq 1.$$

Rationalising the expression gives

$$x_n = \frac{n}{\sqrt{n^2 + n} + n}.$$

Since

$$n < \sqrt{n^2 + n} < n + 1,$$

we obtain

$$\frac{n}{2n + 1} < x_n < \frac{1}{2}.$$

By the algebraic limit laws,

$$\lim_{n \rightarrow \infty} \frac{n}{2n + 1} = \lim_{n \rightarrow \infty} \frac{1}{2 + 1/n} = \frac{1}{2}.$$

The upper bound is the constant sequence $1/2$, so both bounds converge to $1/2$. Hence the squeeze theorem yields

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \frac{1}{2}.$$

2.4.4. Subsequences.

Definition 2.29. Let (x_n) be a sequence, and let (n_k) be a strictly increasing sequence of natural numbers:

$$n_0 < n_1 < n_2 < \dots$$

The sequence (x_{n_k}) , obtained by selecting the terms of (x_n) whose indices are $n_0, n_1, n_2, \dots$, is called a *subsequence* of (x_n) .

Theorem 2.30. If $x_n \rightarrow x$, then every subsequence of (x_n) converges to x .

Proof. Let (x_{n_k}) be a subsequence of (x_n) , and let $\varepsilon > 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - x| < \varepsilon.$$

Because (n_k) is strictly increasing, $n_k \geq k$ for every $k \in \mathbb{N}$. Thus, whenever $k > N$, we also have $n_k > N$, and hence

$$|x_{n_k} - x| < \varepsilon.$$

Therefore $x_{n_k} \rightarrow x$. Since the subsequence was arbitrary, every subsequence of (x_n) converges to the same limit x . $\square$

2.5. 15.09.2026.

2.5.1. Upper and lower bounds.

Definition 2.31. Let $\emptyset \neq E \subseteq \mathbb{R}$. A number $s \in \mathbb{R}$ is an *upper bound* of E if

$$x \leq s \quad \text{for every } x \in E.$$

If such a number exists, E is *bounded above*.

Definition 2.32. Let $\emptyset \neq E \subseteq \mathbb{R}$. A number $p \in \mathbb{R}$ is a *lower bound* of E if

$$p \leq x \quad \text{for every } x \in E.$$

If such a number exists, E is *bounded below*.

Definition 2.33. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. An upper bound s of E is the *least upper bound*, or *supremum*, of E if $s \leq u$ for every upper bound u of E . We write

$$s = \sup E.$$

Definition 2.34. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded below. A lower bound p of E is the *greatest lower bound*, or *infimum*, of E if $p \geq \ell$ for every lower bound ℓ of E . We write

$$p = \inf E.$$

If E is not bounded above, we write $\sup E = +\infty$. Similarly, if E is not bounded below, we write $\inf E = -\infty$.

2.5.2. The completeness axiom.

Completeness axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Corollary 2.35. Every nonempty subset $E \subseteq \mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.

Proof. Let c be a lower bound of E , and define

$$F = \{-x : x \in E\}.$$

Since $c \leq x$ for every $x \in E$, we have $-x \leq -c$. Thus F is bounded above. It is also nonempty, so the completeness axiom gives $\sup F \in \mathbb{R}$. For every $x \in E$,

$$-x \leq \sup F \leq -c, \quad \text{hence} \quad x \geq -\sup F \geq c.$$

Therefore $-\sup F$ is a lower bound of E . Since the argument applies to every lower bound c of E , it is the greatest lower bound. Consequently,

$$\inf E = -\sup F \in \mathbb{R}.$$

$\square$

2.5.3. Maximum and properties of the supremum.

Definition 2.36. A number s is the *greatest element*, or *maximum*, of a set E if $s \in E$ and $x \leq s$ for every $x \in E$. We write $s = \max E$.

Theorem 2.37. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. If $s = \max E$, then $s = \sup E$.

Proof. Since $x \leq s$ for every $x \in E$, the number s is an upper bound of E . Hence $\sup E \leq s$. On the other hand, $s \in E$, so $s \leq \sup E$. Thus $s = \sup E$. $\square$

Theorem 2.38. The supremum of a nonempty subset of $\mathbb{R}$ that is bounded above is unique.

Proof. Suppose that s and s' are both least upper bounds of E . Since s' is an upper bound and s is a least upper bound, $s \leq s'$. Similarly, since s is an upper bound and s' is a least upper bound, $s' \leq s$. Therefore $s = s'$. $\square$

Theorem 2.39. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. For every $\varepsilon > 0$, there exists $x \in E$ such that

$$\sup E - \varepsilon < x \leq \sup E.$$

Proof. Since $\sup E$ is an upper bound, $x \leq \sup E$ for every $x \in E$. Suppose, for a contradiction, that for some $\varepsilon > 0$,

$$x \leq \sup E - \varepsilon \quad \text{for every } x \in E.$$

Then $\sup E - \varepsilon$ is an upper bound of E , but $\sup E - \varepsilon < \sup E$. This contradicts the fact that $\sup E$ is the least upper bound. Hence the claimed element x exists. $\square$

2.5.4. Monotone sequences.

Definition 2.40. A real sequence (x_n) is

- *increasing* (nondecreasing) if $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$;
- *decreasing* (nonincreasing) if $x_{n+1} \leq x_n$ for every $n \in \mathbb{N}$;
- *strictly increasing* if $x_n < x_{n+1}$ for every $n \in \mathbb{N}$;
- *strictly decreasing* if $x_{n+1} < x_n$ for every $n \in \mathbb{N}$.

A sequence is *monotone* if it is increasing or decreasing, and *strictly monotone* if it is strictly increasing or strictly decreasing.

2.6. 17.09.2026.

2.6.1. Monotone convergence.

Theorem 2.41 (Monotone convergence theorem). Let (x_n) be an increasing sequence of real numbers. If there is $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$, then (x_n) converges, and

$$\lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\} \leq M.$$

Proof. The set $E = \{x_n : n \in \mathbb{N}\}$ is nonempty and bounded above by M . By completeness, $s = \sup E$ exists in $\mathbb{R}$, and $s \leq M$. Let $\varepsilon > 0$. By the approximation property of the supremum, there is an index $N \in \mathbb{N}$ such that

$$s - \varepsilon < x_N \leq s.$$

Since (x_n) is increasing, for every $n > N$,

$$s - \varepsilon < x_N \leq x_n \leq s.$$

Hence $|x_n - s| = s - x_n < \varepsilon$, proving $x_n \rightarrow s$. $\square$

Corollary 2.42 (Decreasing case). *Let (x_n) be a decreasing sequence of real numbers. If there is $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$, then (x_n) converges, and*

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\} \geq m.$$

Consequently, every bounded monotone real sequence converges.

Proof. The sequence $(-x_n)$ is increasing and bounded above by $-m$. By theorem 2.41,

$$-x_n \longrightarrow \sup \{-x_n : n \in \mathbb{N}\}.$$

Multiplying by -1 and using the relation between infima and suprema gives

$$x_n \longrightarrow -\sup \{-x_n : n \in \mathbb{N}\} = \inf \{x_n : n \in \mathbb{N}\} \geq m.$$

A bounded monotone sequence satisfies the hypotheses of either the increasing or the decreasing case, so it converges. $\square$

2.6.2. *Examples of decreasing sequences.* In the following two examples the indices start at $n = 1$.

Example 2.43. Let

$$x_1 = 3, \quad x_n = \frac{n}{n-1} \quad (n \geq 2).$$

Show that (x_n) is strictly decreasing and find its limit.

Solution. First, $x_2 = 2 < 3 = x_1$. For $n \geq 2$, all terms are positive and

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)(n-1)}{n^2} = 1 - \frac{1}{n^2} < 1.$$

Thus $x_{n+1} < x_n$ for every $n \geq 1$. Also $x_1 > 1$, and

$$x_n = 1 + \frac{1}{n-1} > 1 \quad (n \geq 2).$$

Hence the sequence is bounded below and converges by corollary 2.42. We identify its limit in two ways.

Method 1: the squeeze theorem. For $n \geq 2$, we have $n-1 \geq n/2$, so

$$1 < x_n = 1 + \frac{1}{n-1} \leq 1 + \frac{2}{n}.$$

Both bounding sequences tend to 1, so $x_n \rightarrow 1$.

Method 2: the infimum. The number 1 is a lower bound of $E = \{x_n : n \geq 1\}$. If $a > 1$, choose an integer $n > 1 + 1/(a-1)$. Then

$$x_n < a \iff \frac{1}{n-1} < a-1 \iff n > 1 + \frac{1}{a-1}.$$

Thus no $a > 1$ is a lower bound of E . Therefore $\inf E = 1$, and the decreasing case of the monotone convergence theorem yields $\lim_{n \rightarrow \infty} x_n = 1$. $\square$

Example 2.44. Let

$$x_n = \frac{(n+1)^2}{n^2+1}, \quad n \geq 1.$$

Show that (x_n) is decreasing and determine its limit.

Solution. Since every term is positive, we can compare successive terms using their ratio:

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{(n+2)^2(n^2+1)}{((n+1)^2+1)(n+1)^2} \\ &= \frac{n^4+4n^3+5n^2+4n+4}{n^4+4n^3+7n^2+6n+2}. \end{aligned}$$

The denominator exceeds the numerator by

$$2n^2 + 2n - 2 = 2(n^2 + n - 1) > 0 \quad (n \geq 1).$$

Therefore $x_{n+1}/x_n < 1$, so the sequence is strictly decreasing. Moreover,

$$x_n = 1 + \frac{2n}{n^2 + 1} > 1,$$

so it is bounded below and converges to $\inf \{x_n : n \geq 1\}$.

We now show that this infimum is 1. It is already a lower bound. Given $a > 1$, choose an integer $n > \max \{1, 2/(a-1)\}$. Then

$$x_n = 1 + \frac{2n}{n^2 + 1} < 1 + \frac{2}{n} < a.$$

Hence a is not a lower bound, and consequently

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \geq 1\} = 1.$$

Alternatively, $1 < x_n < 1 + 2/n$ gives the same limit directly by the squeeze theorem. $\square$

Remark 2.45. The restriction $n \geq 1$ matters for monotonicity in the second example. If we also define x_0 by the same formula, then $x_0 = 1 < x_1 = 2$, so the full sequence is not decreasing. Its tail starting at $n = 1$ is strictly decreasing, and its limit is still 1.

2.6.3. The Bolzano–Weierstrass theorem.

Theorem 2.46 (Bolzano–Weierstrass). *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Let (x_n) be bounded. Choose $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, and define the *tail suprema*

$$a_k = \sup \{x_n : n \geq k\}, \quad k \in \mathbb{N}.$$

Each tail is nonempty and bounded, so completeness ensures that a_k exists and $m \leq a_k \leq M$. Since

$$\{x_n : n \geq k+1\} \subseteq \{x_n : n \geq k\},$$

the number a_k is an upper bound of the smaller tail as well. Therefore $a_{k+1} \leq a_k$. The sequence (a_k) is decreasing and bounded below, so corollary 2.42 gives a limit

$$a = \lim_{k \rightarrow \infty} a_k \in \mathbb{R}.$$

We construct a subsequence $(x_{n_j})_{j \geq 1}$ recursively. By the approximation property of the supremum, choose $n_1 \geq 1$ such that

$$a_1 - 1 < x_{n_1} \leq a_1.$$

Once n_j has been chosen, apply the same property to the tail starting at $n_j + 1$. There is an index $n_{j+1} \geq n_j + 1$ such that

$$a_{n_j+1} - \frac{1}{j+1} < x_{n_{j+1}} \leq a_{n_j+1}.$$

Thus the indices are strictly increasing, as required for a subsequence.

To verify convergence, observe that $a_{n_j+1} \leq a_{n_j+1}$ and $x_{n_{j+1}} \leq a_{n_j+1}$. Consequently, for every $j \geq 1$,

$$a_{n_j} - \frac{1}{j} < x_{n_j} \leq a_{n_j}.$$

For $j = 1$, the lower inequality follows from $a_{n_1} \leq a_1$; for later indices it follows from the recursive choice above. Since (a_{n_j}) is a subsequence of the convergent sequence (a_k) , it tends to a . Also $1/j \rightarrow 0$, so both bounding sequences tend to a . The squeeze theorem now yields $x_{n_j} \rightarrow a$, proving the claim. $\square$

2.7. 22.09.2026.

2.7.1. Bernoulli's inequality.

Theorem 2.47 (Bernoulli's inequality). *For every $x \in \mathbb{R}$ with $x > -1$ and every $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1+nx.$$

Proof. Fix $x > -1$. The cases $n = 0$ and $n = 1$ are equalities. Suppose that $(1+x)^n \geq 1+nx$ for some $n \geq 1$. Since $1+x > 0$, multiplication preserves the inequality, and

$$\begin{aligned} (1+x)^{n+1} &= (1+x)(1+x)^n \\ &\geq (1+x)(1+nx) \\ &= 1 + (n+1)x + nx^2 \\ &\geq 1 + (n+1)x. \end{aligned}$$

The result follows by induction. If $x \neq 0$, then $nx^2 > 0$ in the induction step, so the inequality is strict for $n \geq 2$. $\square$

2.7.2. A sequence defining the number e . Consider the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n, \quad n \geq 1.$$

Lemma 2.48. *The sequence (x_n) is strictly increasing.*

Proof. All terms are positive. For $n \geq 1$,

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n}\right)^n} \\ &= \left(\frac{1 + \frac{1}{n+1}}{1 + \frac{1}{n}}\right)^{n+1} \left(1 + \frac{1}{n}\right) \\ &= \left(\frac{n(n+2)}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &= \left(1 - \frac{1}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &> \left(1 - \frac{n+1}{(n+1)^2}\right) \frac{n+1}{n} = 1. \end{aligned}$$

The strict inequality follows from theorem 2.47, since $-1 < -1/(n+1)^2 < 0$ and $n+1 \geq 2$. Hence $x_{n+1} > x_n$. $\square$

Lemma 2.49. *The sequence (x_n) is bounded above.*

Proof. We use the auxiliary sequence

$$y_n = \left(1 - \frac{1}{n}\right)^n, \quad n \geq 2.$$

This sequence is positive and strictly increasing. Indeed, by theorem 2.47, for $n \geq 2$,

$$\begin{aligned}\frac{y_{n+1}}{y_n} &= \left(\frac{n^2}{n^2-1}\right)^n \frac{n}{n+1} \\ &= \left(1 + \frac{1}{n^2-1}\right)^n \frac{n}{n+1} \\ &\geq \left(1 + \frac{n}{n^2-1}\right) \frac{n}{n+1} \\ &= 1 + \frac{1}{(n+1)(n^2-1)} > 1.\end{aligned}$$

Consequently, $(1/y_n)$ is decreasing. Also, for $n \geq 2$,

$$\left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n}\right) = 1 - \frac{1}{n^2} < 1.$$

Dividing by $1 - 1/n > 0$ and raising to the power n gives

$$x_n < \left(1 - \frac{1}{n}\right)^{-n} = \frac{1}{y_n} \leq \frac{1}{y_2} = \left(1 - \frac{1}{2}\right)^{-2} = 4.$$

Finally, $x_1 = 2 < 4$, so the bound holds for every $n \geq 1$. $\square$

By theorem 2.41, the increasing, bounded sequence (x_n) converges to a real number.

Definition 2.50. The limit

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

is called *Euler's number* (*Neperin luku* in Finnish).

2.8. 24.09.2026.

2.8.1. Sequences tending to infinity.

Definition 2.51. A real sequence (x_n) *tends to positive infinity* if, for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$x_n > M \quad \text{whenever } n > N.$$

We write $x_n \rightarrow +\infty$, or $\lim_{n \rightarrow \infty} x_n = +\infty$.

Definition 2.52. A real sequence (x_n) *tends to negative infinity* if, for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$x_n < M \quad \text{whenever } n > N.$$

We write $x_n \rightarrow -\infty$, or $\lim_{n \rightarrow \infty} x_n = -\infty$.

Remark 2.53. Neither definition requires the sequence to be monotone. The symbols $+\infty$ and $-\infty$ are not real numbers, so these statements do not mean convergence to a real limit.

Theorem 2.54. *If $x_n \rightarrow +\infty$ or $x_n \rightarrow -\infty$, then (x_n) diverges as a sequence of real numbers.*

Proof. Suppose first that $x_n \rightarrow +\infty$, and assume for a contradiction that $x_n \rightarrow x \in \mathbb{R}$. Taking $M = x + 1$ in the definition of an infinite limit gives $N_1 \in \mathbb{N}$ such that $x_n > x + 1$ whenever $n > N_1$. Convergence to x , with $\varepsilon = 1$, gives $N_2 \in \mathbb{N}$ such that $|x_n - x| < 1$ whenever $n > N_2$. For $n > \max\{N_1, N_2\}$, we would have both $|x_n - x| > 1$ and $|x_n - x| < 1$, a contradiction.

If $x_n \rightarrow -\infty$, the same argument uses $M = x - 1$. Eventually $x_n < x - 1$, hence $|x_n - x| > 1$, again contradicting convergence to x . $\square$

2.8.2. A necessary condition for convergence.

Theorem 2.55. *Let (x_n) be a convergent real sequence. For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|x_n - x_{n+k}| < \varepsilon \quad \text{for all } n > N \text{ and } k \in \mathbb{N}.$$

Proof. Write $x = \lim_{n \rightarrow \infty} x_n$, and let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $|x_m - x| < \varepsilon/2$ for every $m > N$. If $n > N$ and $k \in \mathbb{N}$, then $n+k > N$, so the triangle inequality gives

$$|x_n - x_{n+k}| \leq |x_n - x| + |x_{n+k} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Definition 2.56. A real sequence satisfying the condition in theorem 2.55 is called a *Cauchy sequence*. Equivalently, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_p - x_q| < \varepsilon \quad \text{whenever } p, q > N.$$

Indeed, take $n = \min\{p, q\}$ and $k = |p - q|$ to obtain this form from the preceding one. Conversely, set $p = n$ and $q = n + k$.

2.8.3. Nested closed intervals.

Lemma 2.57 (Nested interval lemma). *Let $[a_n, b_n]$, $n \geq 1$, be nonempty closed intervals such that*

$$[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \quad (n \geq 1), \quad \lim_{n \rightarrow \infty} (b_n - a_n) = 0.$$

Then $\bigcap_{n=1}^{\infty} [a_n, b_n]$ contains exactly one real number.

Proof. The nesting gives

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n.$$

Thus (a_n) is increasing and (b_n) is decreasing. Both sequences lie in $[a_1, b_1]$, so the monotone convergence theorem gives real limits

$$a = \lim_{n \rightarrow \infty} a_n, \quad b = \lim_{n \rightarrow \infty} b_n.$$

Since $a_n \leq b_n$, preservation of inequalities under limits yields $a \leq b$. Monotonicity also gives $a_n \leq a \leq b \leq b_n$ for every n , hence

$$0 \leq b - a \leq b_n - a_n.$$

The right-hand side tends to zero, so $a = b$. In particular, $a \in [a_n, b_n]$ for every n , proving existence of a point in the intersection.

For uniqueness, let $y \neq a$. Since $|y - a| > 0$ and $b_n - a_n \rightarrow 0$, choose n with $b_n - a_n < |y - a|$. If y also belonged to $[a_n, b_n]$, then $a, y \in [a_n, b_n]$ would imply $|y - a| \leq b_n - a_n$, a contradiction. Therefore y does not belong to the intersection, and

$$\bigcap_{n=1}^{\infty} [a_n, b_n] = \{a\}.$$

□

2.8.4. Cauchy's convergence criterion.

Theorem 2.58 (Cauchy's convergence criterion). *A real sequence (x_n) converges if and only if it is a Cauchy sequence: for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|x_n - x_{n+k}| < \varepsilon \quad \text{for all } n > N \text{ and } k \in \mathbb{N}.$$

Proof. Necessity was proved in theorem 2.55. Conversely, suppose that (x_n) satisfies the Cauchy condition.

Step 1: boundedness. Apply the condition with $\varepsilon = 1$, and choose a corresponding $N_0 \in \mathbb{N}$. For every $m \geq N_0 + 1$, take $n = N_0 + 1$ and $k = m - (N_0 + 1)$. Then

$$|x_m - x_{N_0+1}| < 1, \quad |x_m| < |x_{N_0+1}| + 1.$$

The finitely many earlier terms are bounded as well. In particular,

$$C = \max\{|x_0|, \dots, |x_{N_0}|, |x_{N_0+1}| + 1\}$$

satisfies $|x_m| \leq C$ for every $m \in \mathbb{N}$.

Step 2: nested intervals from the tails. Define, for $n \geq 1$,

$$p_n = \inf \{x_m : m \geq n\}, \quad s_n = \sup \{x_m : m \geq n\}.$$

Each tail is nonempty and bounded, so its infimum and supremum exist in $\mathbb{R}$. As the tails shrink, their infima increase and their suprema decrease:

$$p_n \leq p_{n+1} \leq s_{n+1} \leq s_n.$$

Consequently, the nonempty closed intervals $[p_n, s_n]$ are nested.

Step 3: the interval lengths tend to zero. Let $\varepsilon > 0$. By the Cauchy condition, choose N so that $|x_r - x_t| < \varepsilon/3$ whenever $r, t > N$. Fix $n > N$. By the approximation properties of the supremum and infimum, there are indices $r, t \geq n$ such that

$$0 \leq s_n - x_r < \frac{\varepsilon}{3}, \quad 0 \leq x_t - p_n < \frac{\varepsilon}{3}.$$

Since $r, t > N$, we obtain

$$\begin{aligned} 0 \leq s_n - p_n &\leq |s_n - x_r| + |x_r - x_t| + |x_t - p_n| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence $s_n - p_n \rightarrow 0$.

Step 4: convergence to the common point. By lemma 2.57, there is exactly one $x \in \mathbb{R}$ such that

$$\bigcap_{n=1}^{\infty} [p_n, s_n] = \{x\}.$$

Given $\varepsilon > 0$, choose n with $s_n - p_n < \varepsilon$. For every $m \geq n$, both x and x_m belong to $[p_n, s_n]$, so

$$|x_m - x| \leq s_n - p_n < \varepsilon.$$

This proves $x_m \rightarrow x$, completing the sufficiency direction. $\square$

2.9. 29.09.2026.

2.9.1. Limits of functions.

Definition 2.59. Let $f : A \rightarrow \mathbb{R}$, where $A \subseteq \mathbb{R}$, and let $x_0 \in \mathbb{R}$. Suppose that there is $r > 0$ such that

$$(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A.$$

The function f has *limit* $m \in \mathbb{R}$ at x_0 if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - m| < \varepsilon \quad \text{whenever } x \in A \text{ and } 0 < |x - x_0| < \delta.$$

We write $\lim_{x \rightarrow x_0} f(x) = m$.

Example 2.60. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 1$. Then $\lim_{x \rightarrow 1} f(x) = 3$.

Proof. Given $\varepsilon > 0$, choose $\delta = \varepsilon/2$. If $0 < |x - 1| < \delta$, then

$$|f(x) - 3| = |2x - 2| = 2|x - 1| < 2\delta = \varepsilon.$$

$\square$

Example 2.61. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 2x + 1, & x \neq 1, \\ 42, & x = 1. \end{cases}$$

Again, $\lim_{x \rightarrow 1} f(x) = 3$, with exactly the same proof: the condition $0 < |x - 1| < \delta$ excludes $x = 1$. Thus the value of a function at the point does not affect its limit.

2.9.2. Linear estimates and Lipschitz continuity.

Example 2.62. Suppose that, for some $C > 0$, $r > 0$, and $L \in \mathbb{R}$,

$$|f(x) - L| \leq C|x - x_0| \quad \text{whenever } 0 < |x - x_0| < r.$$

Then $\lim_{x \rightarrow x_0} f(x) = L$. Indeed, given $\varepsilon > 0$, choose $\delta = \min\{r, \varepsilon/C\}$. For $0 < |x - x_0| < \delta$, we have

$$|f(x) - L| \leq C|x - x_0| < C\delta \leq \varepsilon.$$

If the estimate holds throughout the domain, we may simply take $\delta = \varepsilon/C$.

Definition 2.63. A function $f : A \rightarrow \mathbb{R}$ is *Lipschitz continuous* if there exists a constant $L \geq 0$ such that

$$|f(x) - f(s)| \leq L|x - s| \quad \text{for all } x, s \in A.$$

Example 2.64. The function $f : [0, 1] \rightarrow \mathbb{R}$, $f(x) = \sqrt{x}$, is not Lipschitz continuous. For every integer $n \geq 1$,

$$\frac{|f(1/n) - f(0)|}{|1/n - 0|} = \frac{\sqrt{1/n}}{1/n} = \sqrt{n} \rightarrow +\infty.$$

A Lipschitz constant would have to bound all these quotients, which is impossible.

2.9.3. Choosing a local bound.

Example 2.65. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Then $\lim_{x \rightarrow 2} f(x) = 4$.

Proof. Factor the difference:

$$|f(x) - 4| = |x + 2||x - 2|.$$

If $|x - 2| < 1$, the triangle inequality gives $|x + 2| \leq |x - 2| + 4 < 5$. Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/5\}$. Whenever $0 < |x - 2| < \delta$, it follows that

$$|f(x) - 4| = |x + 2||x - 2| < 5\delta \leq \varepsilon.$$

□

Example 2.66 (Positivity near a point with positive limit). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and suppose that $\lim_{x \rightarrow x_0} f(x) = m > 0$. Then there exists $r > 0$ such that

$$f(x) > 0 \quad \text{for every } x \in (x_0 - r, x_0 + r) \setminus \{x_0\}.$$

Proof. Since $m > 0$, take $\varepsilon = m$ in the definition of the limit. There exists $\delta > 0$ such that $|f(x) - m| < m$ whenever $0 < |x - x_0| < \delta$. Hence

$$-m < f(x) - m < m, \quad \text{so} \quad 0 < f(x) < 2m.$$

Taking $r = \delta$ proves the claim.

□

2.9.4. *Uniqueness of the limit.*

Theorem 2.67. *The limit of a function at a point, if it exists, is unique.*

Proof. Let $f : A \rightarrow \mathbb{R}$, and suppose that $(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A$ for some $r > 0$. Assume, for a contradiction, that

$$\lim_{x \rightarrow x_0} f(x) = a, \quad \lim_{x \rightarrow x_0} f(x) = b, \quad a \neq b.$$

Set $\varepsilon = |a - b|/2 > 0$. There exist $\delta_1, \delta_2 > 0$ such that

$$0 < |x - x_0| < \delta_1 \implies |f(x) - a| < \varepsilon,$$

$$0 < |x - x_0| < \delta_2 \implies |f(x) - b| < \varepsilon.$$

Let $\delta = \min\{r, \delta_1, \delta_2\} > 0$, and choose $x = x_0 + \delta/2$. Then $x \in A$ and $0 < |x - x_0| < \delta$, so both estimates hold. The triangle inequality now gives

$$|a - b| \leq |f(x) - a| + |f(x) - b| < \varepsilon + \varepsilon = |a - b|,$$

a contradiction. Therefore $a = b$. □

2.10. 01.10.2026.

2.10.1. *Further examples of function limits.*

Example 2.68 (Constant functions). If $f(x) = a$ for every $x \in \mathbb{R}$, then $\lim_{x \rightarrow x_0} f(x) = a$ for every $x_0 \in \mathbb{R}$. Indeed, given $\varepsilon > 0$, choose $\delta = 1$. Then $|f(x) - a| = 0 < \varepsilon$ whenever $0 < |x - x_0| < \delta$.

Example 2.69. For $f(x) = x^3$, we have $\lim_{x \rightarrow 2} f(x) = 8$.

Proof. If $|x - 2| < 1$, then $1 < x < 3$, so

$$|x^2 + 2x + 4| < 9 + 6 + 4 = 19.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/19\}$. For $0 < |x - 2| < \delta$, factor the difference to obtain

$$|x^3 - 8| = |x - 2||x^2 + 2x + 4| < 19|x - 2| < 19\delta \leq \varepsilon.$$

□

2.10.2. *Polynomial division.* Polynomial division lets us factor differences such as $P(x) - P(a)$, which is useful in direct proofs of limits.

Theorem 2.70 (Polynomial division). *Let P, D be polynomials with real coefficients, with $D \neq 0$. There are unique polynomials Q, R such that*

$$P = DQ + R,$$

where either $R = 0$ or $\deg R < \deg D$. We call Q the quotient and R the remainder.

Proof. Start with $Q = 0$ and $R = P$. If $R = 0$ or $\deg R < \deg D$, we are done. Otherwise, write the leading terms of R and D as ax^m and bx^d , respectively. Since $b \neq 0$ and $m \geq d$, replace

$$Q \text{ by } Q + \frac{a}{b}x^{m-d}, \quad R \text{ by } R - \frac{a}{b}x^{m-d}D.$$

The identity $P = DQ + R$ is preserved, and the leading term of R cancels. Thus the remainder becomes zero or its degree strictly decreases. After finitely many steps the required degree condition holds.

For uniqueness, suppose $DQ + R = D\tilde{Q} + \tilde{R}$, with both remainders satisfying the degree condition. Then

$$D(Q - \tilde{Q}) = \tilde{R} - R.$$

If $Q - \tilde{Q} \neq 0$, the left side has degree at least $\deg D$, while the right side is zero or has smaller degree. This is impossible. Hence $Q = \tilde{Q}$ and $R = \tilde{R}$. □

Corollary 2.71 (Remainder and factor theorems). *The remainder when dividing $P(x)$ by $x - a$ is $P(a)$. In particular, $x - a$ is a factor of $P(x)$ if and only if $P(a) = 0$.*

Proof. Polynomial division gives $P(x) = (x - a)Q(x) + r$ with constant remainder r . Substituting $x = a$ gives $r = P(a)$. Thus the remainder vanishes exactly when $P(a) = 0$. $\square$

To divide by hand, arrange terms in descending powers, insert zero coefficients for missing powers, and repeatedly divide the leading terms, multiply the divisor, and subtract.

Example 2.72 (Cancelling a factor in a limit). Consider the limit

$$\lim_{x \rightarrow 2} \frac{x^3 + 5x^2 - 2x - 24}{x - 2}.$$

Both numerator and denominator tend to zero, so direct use of the quotient law is not possible. Divide $P(x) = x^3 + 5x^2 - 2x - 24$ by $x - 2$. The successive quotient terms are x^2 , $7x$, and 12 :

$$\begin{aligned} P(x) - x^2(x - 2) &= 7x^2 - 2x - 24, \\ (7x^2 - 2x - 24) - 7x(x - 2) &= 12x - 24, \\ (12x - 24) - 12(x - 2) &= 0. \end{aligned}$$

Therefore

$$x^3 + 5x^2 - 2x - 24 = (x - 2)(x^2 + 7x + 12).$$

For $x \neq 2$, we can cancel $x - 2$, giving

$$\frac{x^3 + 5x^2 - 2x - 24}{x - 2} = x^2 + 7x + 12.$$

Thus the desired limit is $4 + 14 + 12 = 30$. A direct verification uses

$$|x^2 + 7x + 12 - 30| = |x - 2||x + 9|.$$

If $|x - 2| < 1$, then $|x + 9| < 12$. Given $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/12\}$. For $0 < |x - 2| < \delta$, the difference from 30 is less than $12\delta \leq \varepsilon$. The quotient need not be defined at $x = 2$, because the limit uses only nearby points.

Example 2.73 (Using a remainder to simplify a limit). Consider

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}.$$

Again, both numerator and denominator tend to zero. Dividing $x^3 - 1$ by $x^2 - 1$ gives quotient x and remainder $x - 1$, since

$$(x^3 - 1) - x(x^2 - 1) = x - 1.$$

The remainder has degree $1 < 2$, so division is complete. For $x \neq \pm 1$, factor the denominator and cancel:

$$\frac{x^3 - 1}{x^2 - 1} = x + \frac{x - 1}{x^2 - 1} = x + \frac{x - 1}{(x - 1)(x + 1)} = x + \frac{1}{x + 1}.$$

This suggests the limit $1 + 1/2 = 3/2$. To verify it directly, suppose $0 < |x - 1| < 1/2$. Then $x + 1 > 3/2$, and

$$\begin{aligned} \left| \frac{x^3 - 1}{x^2 - 1} - \frac{3}{2} \right| &= \left| (x - 1) + \frac{1}{x + 1} - \frac{1}{2} \right| \\ &\leq |x - 1| + \frac{|x - 1|}{2|x + 1|} \\ &< 2|x - 1|. \end{aligned}$$

Given $\varepsilon > 0$, take $\delta = \min\{1/2, \varepsilon/2\}$. The displayed estimate is then less than $2\delta \leq \varepsilon$, proving the limit.

Remark 2.74. In both examples, division exposes a common factor that can be cancelled for $x \neq x_0$. Functions that agree throughout a punctured neighbourhood of x_0 have the same limit there: their defining ε - δ estimates are identical after shrinking δ to stay in that neighbourhood.

2.10.3. Limits using factorisation and local estimates.

Example 2.75. For $f(x) = x^3 + 5x^2 - 2x - 24$, we have $\lim_{x \rightarrow 2} f(x) = 0$.

Proof. By example 2.72, $f(x) = (x - 2)(x^2 + 7x + 12)$. If $|x - 2| < 1$, then $1 < x < 3$, and hence

$$|x^2 + 7x + 12| < 9 + 21 + 12 = 42.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/42\}$. If $0 < |x - 2| < \delta$, then

$$|f(x) - 0| < 42|x - 2| < 42\delta \leq \varepsilon.$$

□

Example 2.76 (A piecewise function). Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 2|x|, & x < 0, \\ 7, & x = 0, \\ x^2, & x > 0. \end{cases}$$

Then $\lim_{x \rightarrow 0} f(x) = 0$, even though $f(0) = 7$.

Proof. Given $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/2\}$. If $-\delta < x < 0$, then $|f(x)| = 2|x| < 2\delta \leq \varepsilon$. If $0 < x < \delta$, then $x < 1$, so $|f(x)| = x^2 < x < \delta \leq \varepsilon/2 < \varepsilon$. These cases cover $0 < |x| < \delta$. □

Example 2.77 (Keeping a denominator away from zero). Let $f : (0, 1) \rightarrow \mathbb{R}$, $f(x) = 7/x$. Then $\lim_{x \rightarrow 1/2} f(x) = 14$.

Proof. We have

$$\left| \frac{7}{x} - 14 \right| = \frac{14}{|x|} \left| x - \frac{1}{2} \right|.$$

If $|x - 1/2| < 1/4$, then $x > 1/4$, so $14/|x| < 56$. Given $\varepsilon > 0$, choose $\delta = \min\{1/4, \varepsilon/56\}$. Then

$$0 < \left| x - \frac{1}{2} \right| < \delta \implies |f(x) - 14| < 56 \left| x - \frac{1}{2} \right| < 56\delta \leq \varepsilon.$$

□

2.10.4. *A necessary Cauchy condition for function limits.* In the following results, domains contain a punctured neighbourhood of x_0 : for some $r > 0$, $(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A$, as in the previous lecture's definition. All points at which a function is evaluated are understood to belong to its domain.

Theorem 2.78. If $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon$$

whenever $0 < |x - x_0| < \delta$ and $0 < |y - x_0| < \delta$.

Proof. Choose $\delta > 0$ so that $|f(t) - L| < \varepsilon/2$ whenever $0 < |t - x_0| < \delta$. For the stated x, y ,

$$|f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Example 2.79 (A jump prevents a limit). Let $f(x) = 1$ for $x \leq 0$, and $f(x) = 2$ for $x > 0$. The limit at 0 does not exist. Indeed, fix $\varepsilon = 1/2$. For every $\delta > 0$, the points $x = \delta/2$ and $y = -\delta/2$ satisfy $0 < |x|, |y| < \delta$, but

$$|f(x) - f(y)| = 1 > \varepsilon.$$

This contradicts the necessary condition in theorem 2.78.

2.10.5. The sequential criterion for function limits.

Theorem 2.80 (Sequential criterion). *Let $f : A \rightarrow \mathbb{R}$ and $m \in \mathbb{R}$. The following are equivalent:*

- (a) $\lim_{x \rightarrow x_0} f(x) = m$.
- (b) For every sequence (y_n) in $A \setminus \{x_0\}$ with $y_n \rightarrow x_0$, we have $f(y_n) \rightarrow m$.

Proof. Suppose (a) holds, and let $y_n \rightarrow x_0$ with $y_n \in A \setminus \{x_0\}$. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies |f(x) - m| < \varepsilon.$$

There is $N \in \mathbb{N}$ such that $|y_n - x_0| < \delta$ for $n > N$. Since $y_n \neq x_0$, it follows that $|f(y_n) - m| < \varepsilon$ for $n > N$. Thus (b) holds.

Conversely, suppose (b) holds but (a) fails. The negation of (a) says that there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ we can find $y \in A$ satisfying

$$0 < |y - x_0| < \delta, \quad |f(y) - m| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose such a point y_n with $\delta = \min\{r, 1/(n+1)\}$. Then $y_n \neq x_0$ and $|y_n - x_0| < 1/(n+1)$, so $y_n \rightarrow x_0$. But $|f(y_n) - m| \geq \varepsilon_0$ for every n , contradicting (b). Therefore (a) holds. $\square$

Remark 2.81. In the converse proof, we negate the statement that the limit equals the specified number m . This includes both a nonexistent limit and a limit different from m . The restriction $y_n \neq x_0$ is essential because $f(x_0)$ need not equal the limit.

2.10.6. Local boundedness and preservation of bounds.

Theorem 2.82 (Local boundedness). *If $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$, there are $M, \delta > 0$ such that $|f(x)| < M$ whenever $0 < |x - x_0| < \delta$.*

Proof. Choose $\delta > 0$ using $\varepsilon = 1$ in the limit definition. Then

$$|f(x)| \leq |f(x) - L| + |L| < 1 + |L|$$

whenever $0 < |x - x_0| < \delta$. Take $M = 1 + |L| > 0$. $\square$

Theorem 2.83 (Preservation of bounds). *Suppose $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$. If $f(x) \leq M$ throughout some punctured neighbourhood of x_0 , then $L \leq M$. If $f(x) \geq m$ throughout such a neighbourhood, then $L \geq m$. In particular, these conclusions hold if the bounds hold on all of A .*

Proof. For the upper bound, suppose $L > M$ and choose $\varepsilon = L - M > 0$. The limit definition gives $\delta > 0$ such that

$$f(x) > L - \varepsilon = M \quad (0 < |x - x_0| < \delta).$$

Shrink δ if necessary so that this neighbourhood lies in A and the assumed bound holds there. Taking $x = x_0 + \delta/2$ contradicts $f(x) \leq M$. Hence $L \leq M$.

For the lower bound, suppose $L < m$ and take $\varepsilon = m - L > 0$. For all sufficiently close $x \neq x_0$,

$$f(x) < L + \varepsilon = m.$$

Choosing such a point in the neighbourhood where $f(x) \geq m$ gives a contradiction. Thus $L \geq m$. $\square$

2.10.7. *Arithmetic laws for function limits.*

Theorem 2.84. *Let $f, g : A \rightarrow \mathbb{R}$, with $\lim_{x \rightarrow x_0} f(x) = a$ and $\lim_{x \rightarrow x_0} g(x) = b$. Then, for every $c \in \mathbb{R}$,*

$$\begin{aligned}\lim_{x \rightarrow x_0} (f(x) + g(x)) &= a + b, \\ \lim_{x \rightarrow x_0} cf(x) &= ca, \\ \lim_{x \rightarrow x_0} f(x)g(x) &= ab.\end{aligned}$$

If $b \neq 0$, then $g(x) \neq 0$ in some punctured neighbourhood of x_0 , and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{a}{b}.$$

The quotient is defined on $\{x \in A : g(x) \neq 0\}$.

Proof. Let $\varepsilon > 0$. All estimates below apply to points in a sufficiently small punctured neighbourhood of x_0 .

Sum. Choose $\delta_f, \delta_g > 0$ so that $|f(x) - a| < \varepsilon/2$ for $0 < |x - x_0| < \delta_f$, and $|g(x) - b| < \varepsilon/2$ for $0 < |x - x_0| < \delta_g$. With $\delta = \min\{\delta_f, \delta_g\}$,

$$|f(x) + g(x) - (a + b)| \leq |f(x) - a| + |g(x) - b| < \varepsilon.$$

Scalar multiple. If $c = 0$, the result follows from the constant-function example. If $c \neq 0$, choose $\delta > 0$ so that $|f(x) - a| < \varepsilon/|c|$. Then

$$|cf(x) - ca| = |c||f(x) - a| < \varepsilon.$$

Product. By theorem 2.82, there are $M > 0$ and $\delta_0 > 0$ such that $|f(x)| < M$ when $0 < |x - x_0| < \delta_0$. Choose $\delta_1, \delta_2 > 0$ to ensure respectively that

$$|g(x) - b| < \frac{\varepsilon}{2M}, \quad |f(x) - a| < \frac{\varepsilon}{2(|b| + 1)}.$$

For $0 < |x - x_0| < \min\{\delta_0, \delta_1, \delta_2\}$,

$$\begin{aligned}|f(x)g(x) - ab| &= |f(x)(g(x) - b) + b(f(x) - a)| \\ &\leq |f(x)||g(x) - b| + |b||f(x) - a| \\ &< \frac{\varepsilon}{2} + \frac{|b|\varepsilon}{2(|b| + 1)} < \varepsilon.\end{aligned}$$

Quotient. Assume $b \neq 0$. First choose $\delta_0 > 0$ such that $|g(x) - b| < |b|/2$. The reverse triangle inequality gives

$$|g(x)| \geq |b| - |g(x) - b| > \frac{|b|}{2} > 0.$$

Thus the quotient is defined near x_0 . To prove $1/g(x) \rightarrow 1/b$, choose $\delta_1 > 0$ so that $|g(x) - b| < \varepsilon|b|^2/2$. For $0 < |x - x_0| < \min\{\delta_0, \delta_1\}$,

$$\left| \frac{1}{g(x)} - \frac{1}{b} \right| = \frac{|g(x) - b|}{|g(x)||b|} \leq \frac{2|g(x) - b|}{|b|^2} < \varepsilon.$$

The product law applied to f and $1/g$ now yields $f(x)/g(x) \rightarrow a/b$. $\square$

Corollary 2.85 (Polynomial and rational limits). *For every real polynomial P and every $x_0 \in \mathbb{R}$, $\lim_{x \rightarrow x_0} P(x) = P(x_0)$. If Q is another polynomial with $Q(x_0) \neq 0$, then*

$$\lim_{x \rightarrow x_0} \frac{P(x)}{Q(x)} = \frac{P(x_0)}{Q(x_0)}.$$

Proof. The identity function has limit x_0 at x_0 : choose $\delta = \varepsilon$. Repeated use of the product law gives $x^k \rightarrow x_0^k$ for each positive integer k . Constant functions, scalar multiples, and finite sums then give the polynomial statement. The rational statement follows from the quotient law. $\square$

2.11. 06.10.2026.

2.11.1. *Infinite limits at a finite point.* As before, assume that the domain of $f : A \rightarrow \mathbb{R}$ contains a punctured neighbourhood of $x_0 \in \mathbb{R}$.

Definition 2.86. We write $\lim_{x \rightarrow x_0} f(x) = +\infty$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies f(x) > M.$$

Similarly, $\lim_{x \rightarrow x_0} f(x) = -\infty$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies f(x) < M.$$

Example 2.87. For $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f(x) = 1/|x|$, we have $\lim_{x \rightarrow 0} f(x) = +\infty$. If $M \leq 0$, every $x \neq 0$ satisfies $f(x) > M$. If $M > 0$, choose $\delta = 1/M$. Then $0 < |x| < \delta$ implies $f(x) = 1/|x| > 1/\delta = M$.

2.11.2. *One-sided limits.* For the following definitions, the domain A need only contain the corresponding one-sided interval near x_0 .

Definition 2.88. The function $f : A \rightarrow \mathbb{R}$ has *left-hand limit* $L \in \mathbb{R}$ at x_0 , written $\lim_{x \rightarrow x_0^-} f(x) = L$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon.$$

The *right-hand limit* $\lim_{x \rightarrow x_0^+} f(x) = L$ is defined by requiring

$$x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon.$$

The left-hand limit is also denoted by $\lim_{x \rightarrow x_0-} f(x)$, and similarly on the right.

Example 2.89. Let

$$f(x) = \begin{cases} 1 - x, & x \leq 0, \\ 2x - 1, & x > 0. \end{cases}$$

For $x < 0$, $|f(x) - 1| = |x|$, so choosing $\delta = \varepsilon$ proves $\lim_{x \rightarrow 0^-} f(x) = 1$. For $x > 0$, $|f(x) - (-1)| = 2|x|$, so choosing $\delta = \varepsilon/2$ proves $\lim_{x \rightarrow 0^+} f(x) = -1$. The two-sided limit at 0 therefore does not exist.

Theorem 2.90. Suppose A contains a punctured neighbourhood of x_0 . Then $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$ if and only if

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = L.$$

Proof. If the two-sided limit is L , its δ for a given $\varepsilon > 0$ works on each side of x_0 .

Conversely, suppose both one-sided limits equal L . Given $\varepsilon > 0$, choose δ_- and δ_+ from the respective definitions, and put $\delta = \min\{\delta_-, \delta_+\}$. If $0 < |x - x_0| < \delta$, then x is either in $(x_0 - \delta_-, x_0)$ or in $(x_0, x_0 + \delta_+)$. In either case $|f(x) - L| < \varepsilon$. $\square$

Example 2.91. For $f : [0, \infty) \rightarrow \mathbb{R}$, $f(x) = \sqrt{x}$, we have $\lim_{x \rightarrow 0^+} f(x) = 0$. Given $\varepsilon > 0$, take $\delta = \varepsilon^2$. Then $0 < x < \delta$ implies $|f(x) - 0| = \sqrt{x} < \sqrt{\delta} = \varepsilon$. Here only the right-hand limit is defined, since the domain has no points immediately to the left of 0.

2.11.3. *A necessary Cauchy condition for one-sided limits.*

Theorem 2.92. *Suppose A contains an interval (x_0, x_0+r) for some $r > 0$. If $\lim_{x \rightarrow x_0^+} f(x) = L \in \mathbb{R}$, then for every $\varepsilon > 0$ there is $\delta > 0$ such that*

$$|f(x) - f(y)| < \varepsilon \quad \text{whenever } x, y \in (x_0, x_0 + \delta).$$

The analogous statement holds for a left-hand limit.

Proof. Choose $0 < \delta \leq r$ so that $|f(t) - L| < \varepsilon/2$ for all $t \in (x_0, x_0 + \delta)$. Then, for any x, y in this interval,

$$|f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The left-hand case is identical. □